

Technical Report: Vision-Language Domain Transfer via Unified Optimal Flow Matching JEPA (OFM-JEPA)

Revision 2: Verified and Corrected Failure-Mode Analysis

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Abstract

This is a corrected revision of the OFM-JEPA technical report. The original report’s core generative-modeling framework – a dual-track design pairing Discrete Flow Matching for closed-vocabulary VQA with Continuous Reflow-JEPA for open-ended captioning – is sound and well-motivated. However, three of the four “fatal failure modes” used to motivate it rest on *misapplied or unverifiable* theorems: Kirszbraum’s extension theorem, Brouwer’s invariance of domain, and an unreferenced Grönwall bound with an undefined free parameter. We verify each claim against its cited source, show two of the three are direct misapplications (the cited theorems do not say what is claimed), replace the third’s fabricated cubic-exponential rate with the correct, directly-computed linear rate (matching the well-documented score-singularity literature for generative flows near lower-dimensional targets), and correct one architectural bug (a cross-modal EMA update that tracks the wrong encoder). The corrected analysis shows continuous-to-atomic transport is not merely survivable but has an explicit, rigorously well-understood solution via Brenier’s theorem and semi-discrete optimal transport – consistent with, and cross-referenced to, the companion Reflow-JEPA v3 derivation. Discrete Flow Matching remains a valid and recommended architectural choice for closed-vocabulary VQA on its own merits, not because continuous flow matching is otherwise catastrophically unstable.

Original Claim	Verdict	Finding
Lemma 1 (off-manifold L^2 collapse)	[VALID]	Standard, correctly proved conditional-expectation argument.
Failure 1: Kirszbraum Non-Extension	[INVALID CITATION]	Kirszbraum’s theorem is a Lipschitz <i>extension</i> existence result; it says nothing about approximating discontinuous targets. Real phenomenon exists but needs no such citation.
Failure 2: Ambrosio Dimension Collapse	[INVALID CITATION]	Brouwer’s invariance of domain requires <i>injectivity</i> ; the relevant map is not injective, so the theorem does not apply. Ambrosio/DiPerna–Lions’ divergence bound is a sufficient (not necessary) condition for <i>rough</i> fields; irrelevant to an explicit smooth solution. Conclusion is false: an explicit, rigorous solution exists (Brenier 1991).
Failure 3: Super-Exponential Grönwall Instability	[UNVERIFIED / OVER-STATED]	Formula uses undefined σ_i . Correct, directly-computed rate is $O(1/(1-\tau))$ (linear), giving a <i>logarithmic</i> , not exponential, error bound under early stopping.
Failure 4: Representation & Operational Failures	[VALID]	Real, standard, correctly diagnosed. One bug: EMA update direction is cross-modal (nonsensical); corrected below.
Mitigation 1: Discrete Flow Matching	[VALID] (mechanism imprecise)	Gat et al. (2024) is real; its actual formalism is CTMC/generator-based, not literally simplex-ODE as glossed. Legitimate choice on its own merits.
Mitigation 2: Low-dim continuous M_T , $d_{\text{int}} \approx 7\text{--}15$	[UNVERIFIED / OVER-STATED]	Tulchinskii et al. (2023) is real but measures sentence-level word-embedding dimension for AI-text detection, not VQA answer-embedding dimension for this design’s encoder. Needs direct measurement.
Mitigation 3: Norm normalization, EMA, time embeddings, caching, hubness fix	[VALID] (one bug fixed)	Standard and correctly cited (Radovanović et al. 2010; Schnitzer et al. 2012), except EMA direction (fixed below).

1 Setup & Corrected Mathematical Post-Mortem of Failure Modes

Let $\mathcal{Z} \subset \mathbb{R}^d$ be a shared latent space.

- $z_i = E_I(\text{image}) \sim P_I$ is the visual embedding on $M_I \subset \mathcal{Z}$.
- $z_t = E_T(\text{text}) \sim P_T$ is the textual embedding on $M_T \subset \mathcal{Z}$.
- $c \sim P_C$ is a prompt or task conditioning vector.

Assumption 1 (Standing Integrability).

$$\mathbb{E} [\|z_i\|^2 + \|z_t\|^2 \mid c] < \infty \quad a.s. \quad (1)$$

Lemma 1 (Off-Manifold Collapse of L^2 -Regression Predictors). *The minimizer of the standard JEPA predictor loss $\mathcal{L}_{JEPA}(\theta) = \mathbb{E} [\|f_\theta(z_i, c) - z_t\|^2]$ is the conditional expectation:*

$$f_\theta^*(z_i, c) = \mathbb{E} [z_t \mid z_i, c] \quad (2)$$

If M_T is non-convex, there exists a data distribution with P_T multimodal and a set A of conditioning values (z_i, c) with $P_{I,C}(A) > 0$ such that $f_\theta^*(z_i, c) \notin M_T$ for every $(z_i, c) \in A$.

Proof. For fixed (z_i, c) , $\mathbb{E} [\|f - z_t\|^2 \mid z_i, c]$ is a strictly convex quadratic in f . Its unique stationary point is $f = \mathbb{E}[z_t \mid z_i, c]$. If $P_T(\cdot \mid z_i, c) = (1 - \lambda)\delta_a + \lambda\delta_b$ for distinct text embeddings $a, b \in M_T$ and $\lambda \in (0, 1)$, then $f_\theta^*(z_i, c) = (1 - \lambda)a + \lambda b = z^* \notin M_T$. The predictor outputs a spatial centroid in unpopulated ambient space. (Verified correct – no changes.) $\square$

1.1 Failure Mode 1: Trajectory Crossing (Corrected)

Original Claim 1. “Kirszbraun Non-Extension (Crossing Shocks): ...By Kirszbraun’s Extension Theorem, no continuous or Lipschitz neural network v_θ can match a discontinuous conditional mean field across a jump shock.”

[INVALID CITATION] Kirszbraun’s theorem (1934) states the opposite of what is needed here. It is an *extension* result: for Hilbert spaces H_1, H_2 , a subset $E \subseteq H_1$, and a Lipschitz map $f : E \rightarrow H_2$, there exists an extension $\tilde{f} : H_1 \rightarrow H_2$ with $\tilde{f}|_E = f$ and $\text{Lip}(\tilde{f}) = \text{Lip}(f)$ *exactly* – i.e. Lipschitz maps defined on a subset always extend to the whole space without even increasing their Lipschitz constant. This is a positive, constructive existence theorem about extending *regular* maps; it makes no statement whatsoever about whether a continuous function can approximate a *discontinuous* target, which is the actual question at hand.

Proposition 1 (Correct statement of the trajectory-crossing problem). *Let two straight-line data paths $z_i \rightarrow z_t^{(1)}$ and $z_i \rightarrow z_t^{(2)}$ (or from two distinct sources to two distinct targets) intersect at some $(z', \tau') \in \mathcal{Z} \times (0, 1)$. Then the conditional-mean field $u(z, \tau) := \mathbb{E}[z_t - z_i \mid z_\tau = z, c]$ can fail to be continuous at (z', τ') , since the two paths generically carry different velocities through the same point. No continuous function can equal a discontinuous function pointwise at a discontinuity – this follows directly from the definition of continuity, requiring no specialized theorem.*

This is exactly the phenomenon correctly identified (without invoking Kirszbraun) in Liu, Gong & Liu (2023) as the motivation for Reflow, and in the companion Reflow-JEPA theory document’s Theorem 1 discussion. **Crucially, this problem is not merely patched but *eliminated at the population level*** for the specific construction in §1 below: within a single cell of the optimal-transport partition, all trajectories converge to the *same* atom and never cross, so the crossing problem simply does not arise there.

1.2 Failure Mode 2: Divergence Near the Terminal Time (Corrected)

Original Claim 2. “Ambrosio Dimension Collapse: By Brouwer’s Invariance of Domain, continuous ODE flows strictly preserve topological dimension. Transporting a d -dimensional continuous image measure $P_I \ll \mathcal{L}^d$ to a 0-dimensional discrete text measure...forces... $\int_0^1 \operatorname{div}_z v \, d\tau = -\infty$. This violates the $L_\tau^1 L_z^\infty$ near-incompressibility bound required for unique regular Lagrangian flows.”

[INVALID CITATION] on two independent counts.

(a) *Brouwer’s invariance of domain requires injectivity.* The theorem states: if $U \subseteq \mathbb{R}^n$ is open and $f : U \rightarrow \mathbb{R}^n$ is a *continuous injective* map, then $f(U)$ is open (and f is a homeomorphism onto its image). The corollary used for “dimension preservation” – no continuous injection from a higher- into a lower-dimensional space – likewise requires injectivity. The map realized by a well-constructed flow from a continuous source onto a discrete target is emphatically *not* injective: it sends entire positive-measure regions of $\mathcal{Z}$ to single points. Brouwer’s theorem simply does not apply to non-injective maps, so it cannot be used to forbid this construction.

(b) *The Ambrosio/DiPerna–Lions divergence bound is sufficient, not necessary, and is built for a different regime.* DiPerna & Lions (1989) and Ambrosio (2004) establish existence/uniqueness/stability of *regular Lagrangian flows* for *rough* (merely Sobolev or BV) vector fields, under integrability conditions on the divergence, precisely because classical Cauchy–Lipschitz theory does not apply to such fields. When the vector field *is* smooth on the open interval $[0, 1)$ – as it is for the explicit construction below – classical Picard–Lindelöf theory (already correctly invoked as the base theory’s own Theorem 1) suffices, and the DiPerna–Lions/Ambrosio machinery is neither needed nor informative. A theorem’s hypotheses failing does not imply its conclusion is false; it only means that particular theorem is silent.

Proposition 2 (The divergence blow-up is real, benign, and already resolved). *Let Z_0 be an absolutely continuous source (e.g. Gaussian) and $P_{Z_1} = \sum_i m_i \delta_{a_i}$ a discrete target. By Brenier’s theorem (1991), there exists a convex φ such that $T := \nabla \varphi$ pushes Z_0 forward to P_{Z_1} , and T is the a.e.-unique quadratic-cost optimal transport map. For a discrete target, the semi-discrete optimal transport literature (Aurenhammer, 1987; Aurenhammer, Hoffmann & Aronov, 1998) shows T partitions $\mathcal{Z}$ into Laguerre cells Lag_i , each mapped entirely onto its atom a_i ; pairwise cell boundaries have Lebesgue measure zero. Within a single cell, the realized flow is $z_\tau = (1 - \tau)z_0 + \tau a_i$, an explicit affine ODE solution with velocity $v(z, \tau) = (a_i - z)/(1 - \tau)$ and*

$$\operatorname{div}_z v(z, \tau) = -\frac{d}{1 - \tau} \longrightarrow -\infty \quad \text{as } \tau \rightarrow 1^-. \quad (3)$$

The divergence does blow up, exactly as the original claim observed – but this is the expected, benign signature of transporting onto any target of strictly lower dimension than the ambient space, and is not specific to atomic targets or to VQA. It is the exact flow-matching analogue of the well-documented score-function singularity near a data manifold in diffusion models: the score norm is known to scale as $O(1/t)$ near $t \rightarrow 0$ for the VPSDE (Lu et al., cited in Liu, Zhang & Li, 2026), with the singularity decomposing into a bounded tangential component and a divergent $O(1/\sigma)$ normal component (see the tangential/normal decomposition literature, e.g. the analysis surveyed in Liu, Zhang & Li, 2026, §7). Standard, well-understood mitigations – early stopping at $\tau = 1 - \delta$, or EDM-style preconditioning (Karras et al., 2022) that remains well-behaved as the boundary is approached – handle this routinely; it does not require abandoning continuous flow matching.

Remark 1 (Cross-reference). *This construction, and the full proof that it is well-posed almost everywhere and recovers correct multi-answer statistics, is worked out in complete detail (with the explicit Gaussian-source closed form for the two-atom case) in the companion document Reflow-JEPA v3: A Stochastic-Source Correction for Conditional Flow-Matching VQA*

Decoders. *The two documents describe the same underlying mathematical fact from different angles; this report’s Failure Mode 2, correctly resolved, is that document’s Theorem 2.*

1.3 Failure Mode 3: Terminal-Time Sensitivity (Corrected Rate)

Original Claim 3. “Super-Exponential Grönwall Instability: ...the mixture velocity Jacobian norm scales as $\|J_{\text{mixture}}\|_{\text{op}} \geq \frac{\Delta^2}{\sigma_i^2(1-\tau)^3}$. Integrating $L(\tau) \sim (1-\tau)^{-3}$ over $[0, 1-\delta]$ yields a Grönwall error factor of $\exp(1/2\delta^2)$.”

[UNVERIFIED / OVERSTATED] Unverifiable as stated, and the corrected rate is far milder. The symbol σ_i is never defined anywhere in the source document – no Gaussian smoothing, bandwidth, or variance parameter is introduced for Track 2’s construction, so the formula cannot be checked or reproduced. Working directly from the explicit construction of Proposition 2 instead:

Proposition 3 (Correct terminal-time sensitivity rate). *For the field $v(z, \tau) = (a - z)/(1 - \tau)$, the Jacobian is $\nabla_z v(z, \tau) = -I_d/(1 - \tau)$, so*

$$\|\nabla_z v(z, \tau)\|_{\text{op}} = \frac{1}{1 - \tau}, \quad (\text{linear, not cubic, in } (1 - \tau)^{-1}).$$

Consequently, for an early-stopping cutoff at $\tau = 1 - \delta$,

$$\int_0^{1-\delta} \|\nabla_z v(z, \tau)\|_{\text{op}} d\tau = \int_0^{1-\delta} \frac{d\tau}{1 - \tau} = -\ln \delta,$$

a logarithmic, not super-exponential, growth as $\delta \rightarrow 0$. A Grönwall-type bound on integration error over $[0, 1 - \delta]$ therefore grows like δ^{-c} for some modest constant c (polynomially in $1/\delta$, from $\exp(-\ln \delta) = 1/\delta$, not $\exp(1/2\delta^2)$).

This matches the correctly-cited diffusion literature’s $O(1/t)$ or $O(1/\sigma)$ score-singularity rate exactly (a single inverse power, not a cube), and its correctly-integrated consequence is mild and standard: keep a small, non-vanishing stopping margin, exactly as real diffusion/flow-matching samplers already do. The original claim’s practical implication – that continuous CFM is so fragile near $\tau = 1$ that only a wholly separate discrete-state architecture can be trusted for closed-vocabulary VQA – does not follow from the corrected math.

1.4 Failure Mode 4: Representation & Operational Failures (Valid, One Bug Fixed)

- *Representation Collapse:* **[VALID]**. Self-supervised encoders collapse to constant outputs without an anti-collapse mechanism – standard, well-documented (Grill et al., 2020; Bardes, Ponce & LeCun, 2022; Assran et al., 2023).
- *Norm Discrepancy:* **[VALID]**. $\|z_i\| \gg \|z_t\|$ skewing $v = z_t - z_i$ is a real, standard practical issue; L^2 -normalization is the correct standard fix.
- *High-Dimensional Hubness:* **[VALID]**. Confirmed real and correctly attributed: Radovanović, Nanopoulos & Ivanović (2010) establish the phenomenon; Schnitzer, Flexer, Schedl & Widmer (2012) establish Mutual Proximity as a correction. Both citations check out exactly as used.

Original Claim 4. “EMA Target Encoder & Stop-Gradients: Update E_T via Exponential Moving Average (EMA) of E_I with stop-gradients to prevent representation collapse.”

Bug: this describes updating the *text* encoder as an EMA of the *image* encoder – a cross-modal update between two networks operating on different input types, which is not what any cited precedent does and has no clear justification. **[CORRECTED]** In I-JEPA (Assran et al., 2023), BYOL (Grill et al., 2020), and DINO (Caron et al., 2021), a *target* copy of a network is updated as the EMA of the *same* network’s *online* copy, both operating on the same modality. The corrected rule: maintain an online text pipeline (E_T, g_T) trained by gradient descent, and a target copy (E'_T, g'_T) whose weights are $\theta' \leftarrow \eta\theta' + (1 - \eta)\theta$ (same-modality EMA, η close to 1), with z_t computed via the stop-gradient target copy during training.

2 Corrected Theoretical & Algorithmic Mitigations

Mitigation 1: Discrete Flow Matching (DFM) for Closed VQA (Track 1) – Valid, Description Refined

Gat et al. (2024), *Discrete Flow Matching* (NeurIPS 2024), is confirmed as a real, peer-reviewed framework. Its actual mechanism, precisely: rather than a naive ODE on the probability simplex, DFM defines a family of probability paths $p_t(x)$ over a genuinely discrete state space (e.g. per-position categorical variables), evolved via a continuous-time Markov chain (CTMC) with a learned rate/generator matrix, whose marginal-generating velocity has the same functional form as continuous Flow Matching’s generating velocity. Samples remain discrete at every intermediate time; only the probability mass evolves continuously. (The simplex-relaxation framing used in the original report is a reasonable simplification, but is more precisely the mechanism of the separately-cited Dirichlet Flow Matching of Stark et al., 2024, which *does* operate via a literal simplex/Dirichlet construction – the two are related but distinct approaches, both legitimately citable here.)

- **No crossing shocks, by construction:** discrete-state CTMC paths have no continuous-space geometry in which two trajectories can “cross,” sidestepping Proposition 1 entirely rather than requiring Reflow to remove crossings after the fact.
- **Native multi-answer support:** at $t = 1$, DFM’s output distribution over the vocabulary directly represents ambiguity, e.g. $p(\text{“red”}) = 0.5$, $p(\text{“scarlet”}) = 0.5$, matching Lemma 1’s multimodal case without needing the source-distribution correction that continuous CFM requires (Remark 1).
- **Recommended for Track 1 on its own merits** – a clean, purpose-built discrete generative mechanism – *not* because continuous CFM is otherwise unstable (per the corrected Section 1.2 above, it is not).

Mitigation 2: Continuous Reflow-JEPA for Open-Ended VL (Track 2) – Valid, Justification Corrected

- **Dimension collapse is not the reason to prefer a low-dimensional M_T .** Proposition 2 shows continuous-to-discrete (or continuous-to-lower-dimensional) transport already has a complete, rigorous solution; a low-dimensional continuous M_T is not required to avoid a fatal instability. It remains a reasonable modeling choice for open-ended captions on independent grounds (smoother interpolation geometry, fewer parameters needed for the predictor to resolve fine text structure).
- **The $d_{\text{int}} \approx 7\text{--}15$ figure needs direct verification.** Tulchinskii et al. (2023) is a real NeurIPS paper; it measures persistent-homology dimension of *static word embeddings within a sentence*, for the purpose of separating human- from AI-generated text, finding values of $\approx 9\text{--}10$ (alphabetic languages) or ≈ 7 (Chinese) for human text. This is a different embedding

(static word-level) and a different corpus (full fluent sentences) from a VQA answer manifold M_T produced by this design’s own E_T on short answer phrases. **Recommendation:** measure d_{int} of M_T directly, using the same persistent-homology estimator, on the actual answer vocabulary and encoder used here, before relying on this number for architectural decisions (e.g. choosing d_{shared}).

- **Reflow Rectification ($K \geq 2$):** **[VALID]**, matches the base theory (Liu, Gong & Liu, 2023) and the companion Reflow-JEPA v3 document’s Theorem/Corollary on Reflow reaching a fixed point within each transport cell.

Mitigation 3: Representation & Algorithmic Safeguards (Corrected)

1. **L_2 -Norm Normalization:** $z_i \leftarrow z_i / \|z_i\|_2$, $z_t \leftarrow z_t / \|z_t\|_2$. **[VALID]**, unchanged.
2. **[CORRECTED] Same-Modality EMA Target Encoder & Stop-Gradients:** maintain a target copy (E'_T, g'_T) as an EMA of the *online text* pipeline (E_T, g_T) – not of E_I – with stop-gradient applied to the target copy’s output, following I-JEPA/BYOL/DINO precedent.
3. **Sinusoidal FiLM / Cross-Attention Time Embeddings:** **[VALID]**, standard DiT/diffusion practice, unchanged.
4. **Deterministic Reflow Trajectory Caching:** **[VALID]**, unchanged.
5. **Mutual Proximity & LLM Soft-Prompt Decoding:** **[VALID]**; Mutual Proximity correctly attributed to Schnitzer et al. (2012), building on the hubness phenomenon established by Radovanović et al. (2010).

3 Modular Pool of Compatible Component Choices

Table 1 is unchanged from the original except the Text Target Encoder row, corrected to reflect same-modality EMA tracking.

4 Inference & Decoding Pipelines

Pipeline 1: Candidate Bank Snap (Atomic Tasks / VQA Track 1)

1. **Candidate Memory Bank:** Pre-compute $M_T = \{E_T(a_1), \dots, E_T(a_N)\} \in \mathbb{R}^{N \times d}$.
2. **Predictor Forward Pass:**
 - *Track 1 (DFM):* Simulate the CTMC probability path p_τ to $\tau = 1$. Read out top-1 or top- k max probability answers directly: $a^* = \arg \max_{a \in \mathcal{A}} p_1(a)$.
 - *Track 2 (Continuous Reflow):* Execute 1-step or 2-step Euler integration $\hat{z}_t = z_i + v_\theta(z_i, 0, c)$, stopping at $\tau = 1 - \delta$ for a small $\delta > 0$ per Proposition 3 rather than integrating to the singular endpoint exactly.
3. **Text Readout:** Snap $\hat{z}_t$ to M_T via Mutual Proximity Cosine Matching:

$$a^* = \arg \max_{a \in \mathcal{A}} \frac{\langle \hat{z}_t, E_T(a) \rangle}{\|\hat{z}_t\|_2 \|E_T(a)\|_2} \quad (4)$$

Table 1: Pool of Compatible Component Choices for OFM-JEPA (corrected).

Component Module		Recommended Option		Alternative Options	Op-	Function & Compatibility Guarantee
Image	Encoder	DINOv2	(ViT-	SigLIP, OpenCLIP-ViT, Patch-JEPA		Produces dense, isotropic visual embeddings $z_i \in M_I$; requires L_2 -normalization.
Text	Target Encoder	Same-modality EMA text pipeline		Frozen CLIP-Text, T5-Encoder, LLaMA Embed		Embeds ground-truth text $a_t \rightarrow z_t \in M_T$. Target copy tracks the <i>online text encoder</i> via EMA (corrected from v1); prevents collapse without cross-modal confusion.
Normalizer Module		Unit	L_2 -Sphere Projection	LayerNorm ($\gamma = 1, \beta = 0$)		Enforces $\ z_i\ _2 = \ z_t\ _2 = 1.0$, preventing velocity scale distortion.
Time Injection		Sinusoidal	Fourier MLP	Random Fourier Features		Maps scalar $\tau \in [0, 1]$ to 256-D embedding, injected via FiLM modulation.
Predictor	Backbone	DiT-Block (Diffusion Transformer)		Depth-Scaled ResNet-MLP, ICNN		Parameterizes velocity field $v_\theta(z_\tau, \tau, c)$ with time and prompt conditioning.
Track 1 Flow Engine (VQA)		Discrete	Flow Matching (DFM)	Dirichlet Flow Matching		CTMC-based probability paths over the answer vocabulary (Gat et al., 2024); simplex-relaxation variant per Stark et al. (2024).
Track 2 Flow Engine (VL)		Continuous Reflow ($K \geq 2$)		Truncated Schrödinger SDE	Bridge	Transports mass along text manifold M_T ; near-terminal divergence handled by early stopping / EDM-style preconditioning, not by architectural avoidance.
Text	Readout (Atomic)	Cosine	Mutual Proximity Snap	Temperature-Scaled Sinkhorn Plan		Snaps continuous $\hat{z}_t$ to Candidate Bank $M_T = E_T(\mathcal{A})$, eliminating high- D hubness (Schnitzer et al., 2012).
Text	Readout (Open-Ended)	LLM	Soft-Prompt Projection	Autoregressive Decoder	T5	Projects $\hat{z}_t$ via linear head $h_{\text{prefix}} = W_{\text{proj}} \hat{z}_t$ into frozen LLaMA/T5.

Pipeline 2: Open-Ended LLM Projection (General VL Track 2)

1. **Predictor Forward Pass:** Integrate continuous flow to $\tau = 1 - \delta$ (or single pass after distillation), yielding $\hat{z}_t \in \mathbb{R}^d$.
2. **Linear Projection:** Pass $\hat{z}_t$ through projection layer W_{proj} to form soft prompt $h_{\text{prefix}} = W_{\text{proj}} \hat{z}_t$.
3. **LLM Generation:** Feed h_{prefix} into frozen LLaMA/T5 to autoregressively generate open-ended text:

$$P(\text{Text}) = \prod_{l=1}^L P(a_l \mid a_{<l}, h_{\text{prefix}}) \quad (5)$$

5 Experimental & Diagnostic Protocol

All model arms share the identical image encoder E_I (DINOv2-L/14) and text target encoder E_T .

Model Arms

1. **Arm (A) Baseline General VL-JEPA:** Standard L^2 -regression loss $\mathcal{L}_{\text{JEPA}}(\theta) = \mathbb{E} \|f_\theta(z_i, c) - z_t\|^2$. Decoded via top-1 retrieval over M_T .
2. **Arm (B) Unconstrained CFM Flow:** Baseline flow matching model on straight-line ODEs ($K = 1$ round).
3. **Arm (C) DFM-JEPA (Track 1):** Discrete Flow Matching on the answer vocabulary for VQA short answers.
4. **Arm (D) Continuous Reflow-JEPA (Track 2):** Continuous Flow Matching retrained over $K = 2$ Reflow rounds for open-ended captions.

Evaluation Metrics

1. **VQA Accuracy:** Top-1 accuracy on VQA v2.0 and GQA test splits (evaluating Track 1).
2. **Captioning Quality:** CIDEr, BLEU-4, and SPICE scores on COCO Captions test split (evaluating Track 2).
3. **Cross-Modal Retrieval:** Image-to-Text (I $\rightarrow$ T) Recall@1, Recall@5, and Recall@10.
4. **Manifold Adherence Rate:** Empirical probability $\mathbb{P}[\text{dist}(\hat{z}_t, M_T) \leq \varepsilon]$ on held-out data before snapping (testing Lemma 1 avoidance).
5. **Straightness & Crossing Logging:** Tracking the combined empirical proxy

$$\widehat{S_k + V_k} = \frac{1}{N_{\text{val}}} \sum_{j=1}^{N_{\text{val}}} \|z_i^{(j)} - \hat{z}_t^{(k-1,j)}\|^2 - \frac{1}{N_{\text{val}}} \sum_{j=1}^{N_{\text{val}}} \|z_i^{(j)} - \hat{z}_t^{(k,j)}\|^2 \quad (6)$$

on a validation split to select k^* .

6. **[CORRECTED] Terminal-Divergence Diagnostic (new):** directly measure $\|\nabla_z v_\theta(z, \tau, c)\|_{\text{op}}$ (or its Frobenius-norm proxy) on held-out trajectories as $\tau \rightarrow 1$, and check it tracks the corrected $O(1/(1 - \tau))$ rate of Proposition 3 rather than something worse – a direct, checkable test of whether the trained network’s terminal behavior matches the theoretical prediction, replacing reliance on the original document’s unverifiable formula.

7. **[CORRECTED] Intrinsic-dimension measurement (new):** before fixing d_{shared} or d_{int} for Track 2, estimate the persistent-homology dimension of M_T directly on this design’s own answer encoder and vocabulary, following Tulchinskii et al.’s (2023) estimator, rather than importing their sentence-embedding figure.

References

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